

Imaging & Radiology Scheduling System (RAD-SCH)

Software Requirements Specification

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Required:

A system is required for scheduling radiology examinations (MRI/CT) that takes into account the duration of each scan and the capacity of the device

Preparation instructions must be sent to the patient when the appointment is created, and the booking cannot be confirmed unless these instructions are provided

In case of multiple requests for the same device, the system must automatically select the best available appointment based on scan duration, request order, and case priority

In the event of a device failure, all affected appointments must be rescheduled, and patients must be notified

Overview

This document contains the following sections:

Introduction and system overview.

Actors and system users

Functional requirements

Non-functional requirements

Workflow description

Constraints

Critical analysis questions related to the project

1. Introduction

1.1 Purpose

1.1. This document presents the Software Requirements Specification (SRS) for the Imaging & Radiology Scheduling System (RAD-SCH). The purpose of this document is to define the functional and non-functional requirements of the system and provide a clear reference for developers, analysts, testers, and project stakeholders

The RAD-SCH module is part of the MediChain healthcare system and is responsible for managing MRI and CT scan appointments, scheduling operations, patient preparation instructions, notification services, and automatic rescheduling processes in case of device failures.

1.2 Scope

The Imaging & Radiology Scheduling System is designed to improve the management of radiology appointments and optimize the use of MRI and CT devices within healthcare facilities

1.2.1.The system will:

Schedule MRI and CT scan appointments.

Manage patient preparation instructions

Automatically select the best available appointment

Handle appointment prioritization based on urgency

Send notifications to patients.

Automatically reschedule appointments in case of machine failure

Prevent overlapping appointments on the same device

1.2.2.The system will NOT:

Perform medical diagnosis

Manage payroll or human resources

Manage non-radiology hospital departments

Replace medical staff decision-making

Store medical imaging files

1.3 System Objectives

Reduce patient waiting time

Improve device utilization

Automate scheduling decisions

Minimize human errors

2. Actors

2.1 Scheduling / Administrative Staff

Cannot confirm appointments without preparation instruction

Creates appointments in the system

Defines patient preparation instructions

2.2 Patient

Receives preparation instructions after appointment creation

Receives notifications in case of schedule changes or device failures

2.3 MediChain System

Automatically selects the best appointment

Compares requests based on: Scan duration, Priority level, Request order

Automatically reschedules affected appointments during device failures

2.4 Notification System

Sends notifications to patients regarding :

- 1.Appointment confirmations
 - 2.Rescheduling updates
 - 3.Device failure alerts
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3. Functional Requirements

3.1 Appointment Scheduling

The system shall allow scheduling MRI and CT scan appointments

3.2 Preparation Instructions Management

The system shall require preparation instructions before confirming any appointment

3.3 Automatic Appointment Selection

The system shall automatically choose the best appointment slot when multiple requests exist for the same device

3.4 Priority-Based Scheduling

The system shall prioritize appointments according to

Case urgency

Request order

Scan duration

3.5 Device Failure Management

The system shall automatically reschedule all affected appointments if a device becomes unavailable

3.6 Notification Management

The system shall notify patients regarding:

Appointment confirmations

Rescheduling operations

Device failures

4. Non-Functional Requirements

4.1 Reliability

The system shall continue operating efficiently even during unexpected device failures through automatic rescheduling mechanisms

4.2 Accuracy

The system shall accurately calculate scan durations and available device capacity

4.3 Performance

The system shall process multiple appointment requests efficiently without significant delay

4.4 Availability

The system shall support continuous scheduling operations during official working hours

4.5 Usability

The system interface shall be simple and easy for scheduling staff to use

4.6 Scalability

The system shall support future expansion by adding additional radiology devices

5. Constraints

- 5.1.Appointments cannot be confirmed without preparation instructions
- 5.2.Preparation instructions must be sent to the patient immediately after appointment creation
- 5.3.Scheduling must remain within official working hours
- 5.4.Device operating capacity must not be exceeded
- 5.5.Appointment conflicts on the same device are not allowed
- 5.6.All affected appointments must be rescheduled in case of device failure
- 5.7.Patients must be notified of any scheduling changes

6. Critical Questions

6.1.How many MRI and CT devices are available in the laboratory?

6.2.What are the official working hours of the radiology department

6.3.What defines device capacity?

6.4.Are emergency cases given higher priority?

6.5.What is meant by request order?

6.6.Can a patient book multiple appointments on the same day?

6.7.Who has permission to modify or cancel appointments?

6.8.What types of radiology devices are available?

6.9.Do preparation instructions differ between MRI and CT scans?

6.10.What happens if all devices become unavailable?

7. Workflow

7.1 Request and Validation Phase

The staff enters the scan request

The system checks scan duration and device availability

7.2 Preparation Instructions Phase

The staff must define preparation instructions.

Appointment confirmation is blocked if instructions are missing.

7.3 Automatic Scheduling Phase

The system compares requests automatically

Priority is given to urgent cases and earlier requests

7.4 Patient Communication Phase

Patients receive appointment details and preparation instructions immediately after confirmation

7.5 Device Failure Management

If a device fails:

The system identifies affected appointment

Appointments are automatically rescheduled

Notifications are sent to patients

8. Definitions

Term	Definition
MRI	Magnetic Resonance Imaging
CT	Computed Tomography
RAD-SCH	Radiology Scheduling Module
Appointment	Scheduled radiology examination
Preparation Instructions	Instructions given to patients before examination
Notification	Alert sent to the patient

9. References

9.1.IEEE 830 Software Requirements Specification Standard.

9.2.MediChain System Documentation

9.3.Software Engineering Course Materials

9.4.UML Modeling References

9.5.Project Analysis Documents prepared by the RAD-SCH Team

10.Conclusion

This document provided a comprehensive specification of the Radiology Imaging Scheduling System. It defined the system scope, requirements, constraints, and overall behavior. The goal of the system is to improve efficiency reduce waiting time,and automate scheduling decision

— The End —